

Docker Lab 2 Report

Student: Tang Yuanzhi

GitHub Repository: <https://github.com/yi-xian0706/Cloud-Computing>

Lab Location: Lab/Lab2

1. Publishing Ports

This task publishes a container port to make the application accessible from the host computer. I ran a container and mapped port 8080 on the host to port 80 inside the container. After running the container, I used the `docker ps` command to check the running status. I opened a browser and visited `localhost:8080` to confirm the service works correctly. Screenshots include the command output and the browser page.

2. Overriding Container Defaults

This task shows how to replace the default command of a container. I stopped the previous container and started a new container with an interactive shell. Inside the container, I listed files and then exited. This step demonstrates that users can control what command a container runs when it starts. Screenshots show the interactive terminal session.

3. Persisting Container Data

This task uses a Docker volume to keep data safe even after a container is deleted. I created a named volume and attached it to an Nginx container. The volume stores web content so that data remains persistent. I verified the volume exists and checked the container is running with the volume mounted. Screenshots include volume creation and container status.

4. Sharing Local Files

This task mounts a local folder into a container so files stay synchronized. I created a local folder named `docker-lab` and made an `index.html` file with a simple message. I ran a container that uses the current local directory as a mounted folder. I visited the browser to confirm the local webpage displays correctly. Screenshots include the local file and the working web page.

5. Multi-container Applications

This task uses Docker Compose to run multiple services together. I created a `docker-compose.yml` file that defines two services: a web service using Nginx and a database service using MySQL. I started the services in the background and checked that both containers are running. The web service uses the local files, and the database uses a fixed password. After testing, I stopped and removed the services. Screenshots include the compose file content and the running services.

6. Cleanup

I stopped all running containers and removed them to free resources. I also used Docker Compose to stop the multi-container application properly.

Summary

I have completed all required steps for Docker Lab 2. I practiced publishing ports, overriding container commands, persisting data with volumes, sharing local files, and deploying multi-container applications. All functions work as expected, and screenshots are included in the lab folder.

```
Microsoft Windows [版本 10.0.26200.8246]
(c) Microsoft Corporation。保留所有权利。

C:\Windows\System32>docker pull docker/getting-started
Using default tag: latest
latest: Pulling from docker/getting-started
ee68d3549ec8: Downloading [=====>] 3.146MB/11.49MB
4f7e34c2de10: Already exists
33e0cbbb4673: Already exists
failed to copy: httpReadSeeker: failed open: failed to do request: Get "https://docker.mirrors.usc.edu.cn/v2/docker/getting-started/blobs/sha256:ee68d3549ec890fe6714463c6bfc4b33e306afc2615a08630331ee41822144c3?ns=docker.io": EOF

C:\Windows\System32>docker run -d -p 8080:80 docker/getting-started
Unable to find image 'docker/getting-started:latest' locally
latest: Pulling from docker/getting-started
4f7e34c2de10: Pull complete
33e0cbbb4673: Pull complete
ee68d3549ec8: Pull complete
Digest: sha256:d79336f4812b6547a53e735480dde67f8f7071b414fbd9297609ffb989abc1
Status: Downloaded newer image for docker/getting-started:latest
3f46352948b4b5d990bef226437c870e588188f1ab79be5d08de9dd2fd71a76e

C:\Windows\System32>docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
3f46352948b4   docker/getting-started   "/docker-entrypoint.…"   About a minute ago   Up About a minute   0.0.0.0:8080->8080/tcp, [::]:8080->80/tcp
frosty_mcnullty
```

Getting Started - x

千问 >

localhost:8080/tutorial/

千问 >

Getting Started

Getting Started

The command you just ran

Congratulations! You have started the container for this tutorial! Let's first explain the command that you just ran. In case you forgot, here's the command:

```
docker run -d -p 80:80 docker/getting-started
```

You'll notice a few flags being used. Here's some more info on them:

- `-d` - run the container in detached mode (in the background)
- `-p 80:80` - map port 80 of the host to port 80 in the container
- `docker/getting-started` - the image to use

Pro tip

You can combine single character flags to shorten the full command. As an example, the command above could be written as:

```
docker run -dp 80:80 docker/getting-started
```

```
in          etc          mnt          run          tmp
dev         home         opt          sbin         usr
docker-entrypoint.d lib        proc         srv          var
docker-entrypoint.sh media      root         sys
# exit

C:\Windows\System32>docker volume create mydata
mydata

C:\Windows\System32>docker run -d -p 8080:80 -v mydata:/usr/share/nginx/html nginx
Unable to find image 'nginx:latest' locally
```

```
管理员: 命令提示符
Microsoft Windows [版本 10.0.26200.8246]
(c) Microsoft Corporation. 保留所有权利。

C:\Windows\System32>docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:6e23479198b998e5e25921dff8455837c7636a67111a04a635cf1bb363d199dc
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest

C:\Windows\System32>docker run -d -p 8080:80 -v mydata:/usr/share/nginx/html nginx
084ab0b5559aa0c9a6dad75f2fa80e1224bef57a567924780f3a8e39af9552f2

C:\Windows\System32>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS
084ab0b5559a   nginx    "/docker-entrypoint. ..." 22 minutes ago Up 22 minutes 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
inspiring_diffie

C:\Windows\System32>
```

Getting Started - x + 千问 > localhost:8080/tutorial/ 翻译 网址导航

Getting Started

Getting Started

The command you just ran

Congratulations! You have started the container for this tutorial! Let's first explain the command that you just ran. In case you forgot, here's the command:

```
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```

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- `docker/getting-started` - the image to use

Pro tip

You can combine single character flags to shorten the full command. As an example, the command above could be written as:

```
docker run -dp 80:80 docker/getting-started
```

Hello Docker! This is my Lab 2

```
time="2026-05-02T02:12:02+08:00" level=warning msg="C:\\Users\\Lenovo\\Desktop\\docker-lab\\docker-compose.
yaml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 0/1
[+] up 0/1mysql:5.7 Pulling
6.5
[+] up 0/1mysql:5.7 Pulling
6.
[+] up 0/1mysql:5.7 Pulling
[+] up 0/1mysql:5.7 Pulling
6
[+] up 16/17ql:5.7 Pulling
7.
✓ Image mysql:5.7 Pulled 158.5s
✓ Network docker-lab_default Created 0.1s
- Container docker-lab-web-1 Starting 1.9s
✓ Container docker-lab-db-1 Started 1.6s
Error response from daemon: failed to set up container networking: driver failed programming external connec
tivity on endpoint docker-lab-web-1 (635b09d77d47fab9b483f593cccb4ccb2edc447c089a6bea55a8ecdb823b566): Bi
nd for 0.0.0.0:8080 failed: port is already allocated

C:\\Users\\Lenovo\\Desktop\\docker-lab>docker-compose ps
time="2026-05-02T02:16:12+08:00" level=warning msg="C:\\Users\\Lenovo\\Desktop\\docker-lab\\docker-compose.
yaml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
NAME IMAGE COMMAND SERVICE CREATED STATUS
PORTS
docker-lab-db-1 mysql:5.7 "docker-entrypoint.s..." db About a minute ago Up About a minute
3306/tcp, 33060/tcp

C:\\Users\\Lenovo\\Desktop\\docker-lab>
```

```
2f291806c6ad7c93feb224396b3d136e127b659cfd8ff7a7ca54569114010439

C:\\Users\\Lenovo\\Desktop\\docker-lab>cd C:\\Users\\Lenovo\\Desktop\\docker-lab

C:\\Users\\Lenovo\\Desktop\\docker-lab>docker-compose up -d
time="2026-05-02T02:12:02+08:00" level=warning msg="C:\\Users\\Lenovo\\Desktop\\docker-lab\\docker-compose.
yaml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 0/1
[+] up 0/1mysql:5.7 Pulling
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[+] up 0/1mysql:5.7 Pulling
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6
[+] up 16/17ql:5.7 Pulling
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✓ Network docker-lab_default Created 0.1s
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tivity on endpoint docker-lab-web-1 (635b09d77d47fab9b483f593cccb4ccb2edc447c089a6bea55a8ecdb823b566): Bi
nd for 0.0.0.0:8080 failed: port is already allocated

C:\\Users\\Lenovo\\Desktop\\docker-lab>docker-compose ps
time="2026-05-02T02:16:12+08:00" level=warning msg="C:\\Users\\Lenovo\\Desktop\\docker-lab\\docker-compose.
yaml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
NAME IMAGE COMMAND SERVICE CREATED STATUS
PORTS
docker-lab-db-1 mysql:5.7 "docker-entrypoint.s..." db About a minute ago Up About a minute
3306/tcp, 33060/tcp

C:\\Users\\Lenovo\\Desktop\\docker-lab>
```

```

docker-lab-db-1    mysql:5.7      "/docker-entrypoint.s..." db      About a minute ago   Up About a minute
3306/tcp, 33060/tcp

C:\Users\Lenovo\Desktop\docker-lab>docker stop 94e8596ff454
94e8596ff454

C:\Users\Lenovo\Desktop\docker-lab>docker-compose down
time="2026-05-02T02:16:59+08:00" level=warning msg="C:\\Users\\Lenovo\\Desktop\\docker-lab\\docker-compose.
yaml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"

[+] down 3/3
✓ Container docker-lab-web-1   Removed      0.0s
✓ Container docker-lab-db-1    Removed      1.5s
✓ Network docker-lab_default   Removed      0.4s

C:\Users\Lenovo\Desktop\docker-lab>docker-compose up -d
time="2026-05-02T02:18:26+08:00" level=warning msg="C:\\Users\\Lenovo\\Desktop\\docker-lab\\docker-compose.
yaml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"

[+] up 3/3
✓ Network docker-lab_default   Created      0.1s
✓ Container docker-lab-db-1    Started      0.6s
✓ Container docker-lab-web-1    Started      0.7s

C:\Users\Lenovo\Desktop\docker-lab>docker-compose ps
time="2026-05-02T02:19:12+08:00" level=warning msg="C:\\Users\\Lenovo\\Desktop\\docker-lab\\docker-compose.
yaml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"

NAME                IMAGE        COMMAND                  SERVICE    CREATED        STATUS        PORTS
docker-lab-db-1     mysql:5.7    "/docker-entrypoint.s..." db         46 seconds ago Up 45 seconds 3306/tcp
, 33060/tcp
docker-lab-web-1    nginx        "/docker-entrypoint...." web        46 seconds ago Up 45 seconds 0.0.0.0
:8081->80/tcp, [::]:8081->80/tcp

C:\Users\Lenovo\Desktop\docker-lab>

```



Hello Docker! This is my Lab 2